

GRAPH THEORY

(1 Marks)

1. GATE CSE 2024 Set 2

MCQ (Single Correct Answer) Marks: +1 / -0.33

Let A be the adjacency matrix of a simple undirected graph G . Suppose A is its own inverse. Which one of the following statements is always TRUE?

- (A) G is a cycle
- (B) G is a perfect matching
- (C) G is a complete graph
- (D) There is no such graph G

2. GATE CSE 2024 Set 1

Numerical Marks: +1 / -0

The number of spanning trees in a complete graph of 4 vertices labelled A, B, C, and D is _____

3. GATE CSE 2022

MCQ (More than One Correct Answer) Marks: +1 / -0

Which of the following statements is/are TRUE for a group G ?

- (A) If for all $x, y \in G$, $(xy)^2 = x^2y^2$, then G is commutative.
- (B) If for all $X \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the identity element of G .
- (C) If the order of G is 2, then G is commutative.
- (D) If G is commutative, then a subgroup of G need not be commutative.

4. GATE CSE 2022

Numerical Marks: +1 / -0

Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is _____.

5. GATE CSE 2021 Set 1

Numerical Marks: +1 / -0

In an undirected connected planar graph G , there are eight vertices and five faces. The number of edges in G is _____

6. GATE CSE 2019

MCQ (Single Correct Answer) Marks: +1 / -0.33

Let G be an undirected complete graph on n vertices, where $n > 2$. Then, the number of different Hamiltonian cycles in G is equal to

- (A) $n!$
- (B) 1
- (C) $(n - 1)!$
- (D) $\frac{(n - 1)!}{2}$

7. GATE CSE 2018

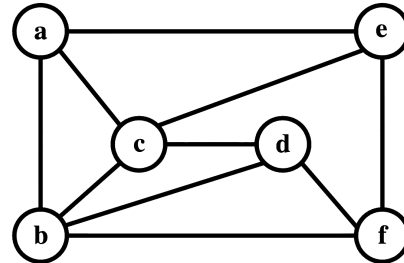
Numerical Marks: +1 / -0

Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____.

8. GATE CSE 2018

Numerical Marks: +1 / -0

The chromatic number of the following graph is _____.



9. GATE CSE 2016 Set 2

Numerical Marks: +1 / -0

The minimum number of colours that is sufficient to vertex-colour any planar graph is _____.

10. GATE CSE 2014 Set 1

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let $G = (V, E)$ be a directed graph where V is the set of vertices and E the set of edges. Then which one of the following graphs has the same strongly connected components as G ?

- (A) $G_1 = (V, E_1)$ where $E_1 = \{(u, v) \notin E\}$
- (B) $G_2 = (V, E_2)$ where $E_2 = \{(u, v) \in E\}$
- (C) $G_3 = (V, E_3)$ where $E_3 = \{(u, v) \mid \text{there is a path of length } \leq 2 \text{ from } u \text{ to } v \text{ in } E\}$
- (D) $G_4 = (V_4, E)$ where V_4 is the set of vertices in G which are not isolated.

11. GATE CSE 2014 Set 1

Numerical Marks: +1 / -0

The maximum number of edges in a bipartite graph on 12 vertices is _____.

12. GATE CSE 2014 Set 3

Numerical Marks: +1 / -0

Let G be a group with 15 elements. Let L be a subgroup of G . It is known that $L \neq G$ and that the size of L is at least 4. The size of L is _____.

13. GATE CSE 2013

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider an undirected random^o graph of eight vertices. The probability that there is an edge between a pair of vertices is $1/2$. What is the expected number of unordered cycles of length three?

- (A) $1/8$
- (B) 1
- (C) 7
- (D) 8

14. GATE CSE 2013

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which of the following statements is/are TRUE for undirected graphs?

P: Number of odd degree vertices is even.

Q: Sum of degrees of all vertices is even.

- (A) P only
- (B) Q only
- (C) Both P and Q
- (D) Neither P and Q

15. GATE CSE 2012

MCQ (Single Correct Answer) Marks: +1 / -0.3

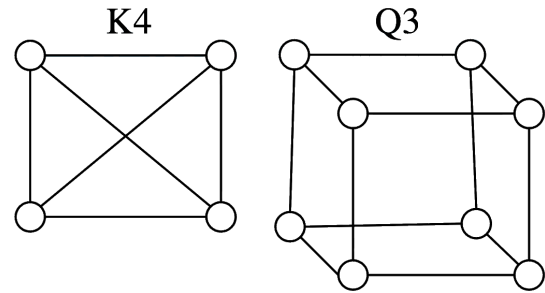
Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of bounded faces in any embedding of G on the plane is equal to

- (A) 3
- (B) 4
- (C) 5
- (D) 6

16. GATE CSE 2011

MCQ (Single Correct Answer) Marks: +1 / -0.3

K_4 and Q_3 are graphs with the following structures.



Which one of the following statements is **TRUE** in relation to these graphs?

- (A) K_4 is planar while Q_3 is not
- (B) Both K_4 and Q_3 are planar
- (C) Q_3 is planar while K_3 is not
- (D) Neither K_4 nor Q_3 is planar

17. GATE CSE 2010

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let $G = (V, E)$ be a graph. Define $\xi(G) = \sum_d i_d \times d$,

where i_d is the number of vertices of degree d in G . If S and T are two different trees with $\xi(S) = \xi(T)$, then

- (A) $|S| = 2|T|$
- (B) $|S| = |T| - 1$
- (C) $|S| = |T|$
- (D) $|S| = |T| + 1$

18. GATE CSE 2009

MCQ (Single Correct Answer) Marks: +1 / -0.3

What is the chromatic number of an n -vertex simple connected graph which does not contain any odd length cycle? Assume $n \geq 2$.

- (A) 2
- (B) 3
- (C) $n - 1$
- (D) n

19. GATE CSE 2009

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which one of the following is TRUE for any simple connected undirected graph with more than 2 vertices?

- (A) No two vertices have the same degree.
- (B) At least two vertices have the same degree.
- (C) At least three vertices have the same degree.
- (D) All vertices have the same degree.

20. GATE CSE 2008

MCQ (Single Correct Answer) Marks: +1 / -0.3

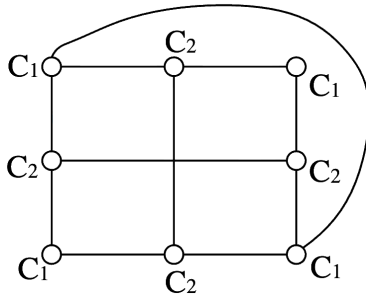
What is the size of the smallest MIS (Maximal Independent Set) of a chain of nine nodes?

- (A) 5
(B) 4
(C) 3
(D) 2

21. GATE CSE 2008

MCQ (Single Correct Answer) Marks: +1 / -0.3

What is the chromatic number of the following graph?



- (A) 2
(B) 3
(C) 4
(D) 5

22. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider a weighted undirected graph with positive edge weights and let uv be an edge in the graph. It is known that the shortest path from the source vertex s to u has weight 53 and the shortest path from s to v has weighted 65. Which one of the following statements is always true?

- (A) $\text{weight}(u, v) < 12$
(B) $\text{weight}(u, v) \leq 12$
(C) $\text{weight}(u, v) > 12$
(D) $\text{weight}(u, v) \geq 12$

23. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +1 / -0.3

The height of a binary tree is the maximum number of edges in any root to leaf path. The maximum number of nodes in a binary tree of height h is:

- (A) $2^h - 1$
(B) $2^{h-1} - 1$
(C) $2^{h+1} - 1$
(D) 2^{h+1}

24. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let G be the non-planar graph with minimum possible number of edges. Then G has

- (A) 9 edges and 5 vertices
(B) 9 edges and 6 vertices
(C) 10 edges and 5 vertices
(D) 10 edges and 6 vertices

25. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +1 / -0.3

The maximum number of binary trees that can be formed with three unlabeled nodes is:

- (A) 1
(B) 5
(C) 4
(D) 3

26. GATE CSE 2006

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider a weighted complete graph G on the vertex set $\{v_1, v_2, \dots, v_n\}$ such that the weight of the edge (v_i, v_j) is $2|i - j|$. The weight of a minimum spanning tree of G is

- (A) $n - 1$
(B) $2n - 2$
(C) $\binom{n}{2}$
(D) n^2

27. GATE CSE 2006

MCQ (Single Correct Answer) Marks: +1 / -0.3

If all the edge weights of an undirected graph are positive, then any subset of edges that connects all the vertices and has minimum total weight is a

- (A) Hamiltonian cycle
(B) grid
(C) hypercube
(D) tree

28. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let G be the simple graph with 20 vertices and 100 edges. The size of the minimum vertex cover of G is 8. Then, the size of the maximum independent set of G is:

- (A) 12
(B) 8
(C) Less than 8
(D) More than 12

29. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let G be a simple connected planar graph with 13 vertices and 19 edges. Then, the number of faces in the planar embedding of the graph is:

- (A) 6
 - (B) 8
 - (C) 9
 - (D) 13
-

30. GATE CSE 2004

MCQ (Single Correct Answer) Marks: +1 / -0.3

What is the maximum number of edges in an acyclic undirected graph with n vertices?

- (A) $n - 1$
 - (B) n
 - (C) $n + 1$
 - (D) $2n - 2$
-

31. GATE CSE 2003

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let G be an arbitrary graph with n nodes and k components. If a vertex is removed from G , the number of components in the resultant graph must necessarily lie between

- (A) k and n
 - (B) $k - 1$ and $k + 1$
 - (C) $k - 1$ and $n - 1$
 - (D) $k + 1$ and $n - k$
-

32. GATE CSE 2002

MCQ (Single Correct Answer) Marks: +1 / -0.3

Maximum number of edges in a n - node undirected graph without self loops is

- (A) n^2
 - (B) $n(n - 1)/2$
 - (C) $n - 1$
 - (D) $(n + 1)(n)/2$
-

33. GATE CSE 1994

MCQ (Single Correct Answer) Marks: +1 / -0.3

The number of distinct simple graph with upto three nodes is

- (A) 15
 - (B) 10
 - (C) 7
 - (D) 9
-

34. GATE CSE 1992

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which of the following is/are tautology?

- (A) $(a \vee b) \rightarrow (b \wedge c)$
 - (B) $(a \wedge b) \rightarrow (b \vee c)$
 - (C) $(a \vee b) \rightarrow (b \rightarrow c)$
 - (D) $(a \rightarrow b) \rightarrow (b \rightarrow c)$
-